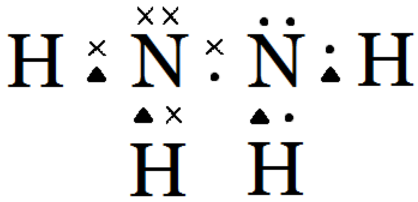
| Question | | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| **8.** | (a) | (i) | | furthest line to the left | 1 | | | 1 | | |
| | | (ii) | | electron falls from higher energy levels to lower energy levels / to n = 2 (1)<br><br>the difference between any two energy levels is fixed / energy levels are quantised (1) | 2 | | | 2 | | |
| | (b) | (i) | | this is the energy required to remove an electron from an atom of hydrogen / for an electron to go from n = 1 to n = $\infty$ (1)<br><br>in the gaseous state (1)<br><br>[award (1) for correct equation with state symbols but no explanation] | 2 | | | 2 | | |
| | | (ii) | | E = hf (1)<br><br>$f = 2.18 \times 10^{-18} / 6.63 \times 10^{-34}$ (1)<br><br>$f = 3.29 \times 10^{15}$ (1) | 1 | 2 | | 3 | 3 | |
| | (c) | | | $V_2 = \dfrac{163 \times 273}{398} = 112$ (1)<br><br>n hydrazine = 112/22400 = 0.00500 (1)<br><br>$M_r$ hydrazine = 0.160/0.005 = 32<br>so molecular formula is $N_2H_4$ (1)<br><br>ecf possible<br><br>credit alternative method using pV = nRT | | 2 | 1 | 3 | 2 | |

| Question | | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| | (d) | (i) | | <br>shared pair of electrons between N atoms (1)<br>rest of electrons correct (1)<br>accept all electrons represented as dots/crosses | | 2 | | 2 | | |
| | | (ii) | | a covalent bond where the electrons are not shared equally between the atoms / unequal electron density | 1 | | | 1 | | |
| | (e) | | | $N_2H_4 + H_2O \rightleftharpoons N_2H_5^+ + OH^-$<br>accept<br>$N_2H_4 + 2H_2O \rightleftharpoons N_2H_6^{2+} + 2OH^-$ | | 1 | | 1 | | |
| | | | | **Question 8 total** | **7** | **7** | **1** | **15** | **5** | **0** |